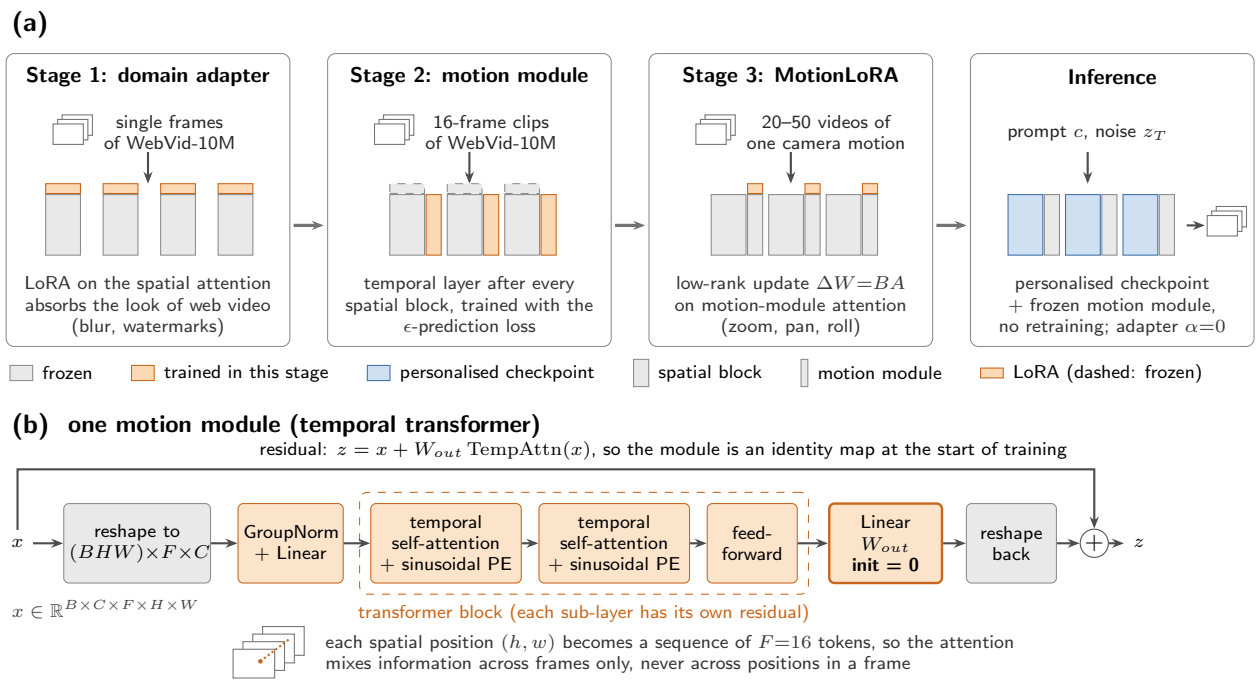

(a)
Stage 1: domain adapter
single frames of WebVid-10M
LoRA on the spatial attention absorbs the look of web video (blur, watermarks)
Stage 2: motion module
16-frame clips of WebVid-10M
temporal layer after every spatial block, trained with the ϵ-prediction loss
Stage 3: MotionLoRA
20–50 videos of one camera motion
low-rank update $\Delta W = BA$ on motion-module attention (zoom, pan, roll)
Inference
prompt c, noise z_T
personalised checkpoint + frozen motion module, no retraining; adapter $\alpha = 0$
frozen
trained in this stage
personalised checkpoint
spatial block
motion module
LoRA (dashed: frozen)
(b) one motion module (temporal transformer)
residual: $z = x + W_{out}\,\mathrm{TempAttn}(x)$, so the module is an identity map at the start of training
x
reshape to $(BHW) \times F \times C$
GroupNorm + Linear
temporal self-attention + sinusoidal PE
temporal self-attention + sinusoidal PE
feed-forward
Linear W_{out} init = 0
reshape back
z
$x \in \mathbb{R}^{B \times C \times F \times H \times W}$
transformer block (each sub-layer has its own residual)
each spatial position (h, w) becomes a sequence of $F = 16$ tokens, so the attention mixes information across frames only, never across positions in a frame